ALICE L WALTON SCHOOL OF MEDICINE

Secondary Hypertension

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Adapted from: Secondary Hypertension

Listen to this Brick

0:00 / 31:41 1x

Learning Objectives (1) Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation, pathophysiology, diagnosis, and management of secondary hypertension caused by the various mentioned disorders.

CASE CONNECTION

JK is a 54-year-old female who comes to the office seeking a second opinion in the management of her hypertension. "My doctor keeps adding medications and increasing the pills I have to take, but my blood pressure is still high." JK currently is prescribed lisinopril, amlodipine, and hydrochlorothiazide—all at the maximum doses—and she has been taking them as directed. During her exam, JK's blood pressure is 160/92 mm Hg. Her physical exam is unremarkable, except for an abdominal bruit.

What do you suspect is the cause of JK's hypertension? How will you explain the diagnosis and management to her? Consider your answers as you read, and we'll revisit JK at the end of the brick.

GO TO CONCLUSION ↓

What Is Secondary Hypertension? (LO #1)

Hypertension is defined as a systolic blood pressure more than 130 mm Hg and/or diastolic blood pressure more than 90 mm Hg on average of at least two separate measurements. Most hypertension (90%- 95%) is primary (essential), meaning that no identifiable disease is responsible for it. But, in the remaining 5% to 10% of patients (**secondary hypertension**), there is an associated underlying cause (Figure 1).

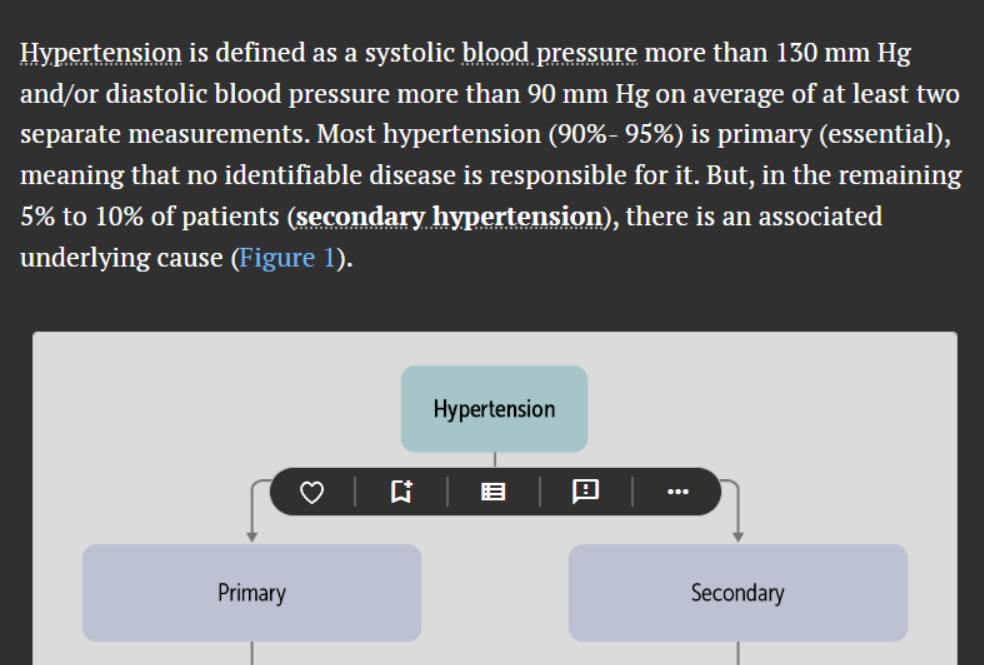
Hypertension
Primary
Secondary
90%-95% of cases
Cause is unknown
Results from complex interaction between genes, lifestyle, and environmental factors
5%-10% of cases
Results from an identifiable cause eg, kidney, renovascular, or endocrine disease; pregnancy, obesity, sleep apnea

Figure 1. Primary versus Secondary Hypertension.

Types of Secondary Hypertension

Possible causes of secondary hypertension include:

- Renal disorders
- Endocrine disease
- Obstructive sleep apnea
- Medications

Mnemonic

Causes of secon ABC²DE: A a, **B**ruits (seen in renal artery stenosis and fibromuscular dysplasia), **C**atecholamines (pheochromocytoma) and **C**hronic kidney disease (CKD), **D**rugs (oral contraceptive pills, NSAIDs, pseudoephedrine, cocaine, caffeine, amphetamines, and steroids), and **E**ndocrine disorders (hyperaldosteronism, hypercortisolism, and thyroid dysfunction)

CLINICAL CORRELATION

An abdominal **bruit** is an abnormal (whooshing or swishing) sound heard when listening to the abdomen with a stethoscope. In a hypertensive patient, an abdominal bruit should raise the immediate consideration of renal artery stenosis.

Clinical Features Suggestive of Secondary Hypertension

Like primary hypertension, secondary hypertension is usually asymptomatic and found on a routine vital sign check. However, there are clinical features that may suggest a secondary cause. Here are some general guidelines:

- New-onset hypertension in patients age younger than 25 years or older than 65 years
- Patients with histories typical of the main secondary causes (see below)
- Patients with bruits or signs of chronic kidney disease (edema)
- Patients with hypokalemia who are not taking diuretics
- Patients present (blood pressure >160/120 mm Hg), including those presenting initially with hypertensive emergency, who have signs of end-organ damage like heart or kidney failure

Epidemiology

Compared to primary hypertension, which is most common in older patients, secondary hypertension exhibits bimodal distribution, being most prevalent at the extremes of age. It accounts for 70% to 85% of hypertension in children younger than age 12 years and approximately 17% of cases in adults older than 64 years.

Secondary hypertension is most common in which age group(s)?

What Are Renal Causes of Secondary Hypertension? (LO #1)

Since the kidney is a key regulator of blood pressure, renal disorders are a common cause of secondary hypertension. The increase in blood pressure results from 1. increased sodium and water retention due to decreased renal excretion and 2. an activated renin-angiotensin-aldosterone system (RAAS), which enhances sodium retention and causes vasoconstriction, increasing systemic vascular resistance.

How do renal diseases cause hypertension?

Chronic Kidney Disease

Chronic kidney disease (CKD) can be a complication or cause of hypertension. Patients are often asymptomatic and may present either with a decreased glomerular filtration rate (GFR) (increased serum blood urea nitrogen [BUN]/creatinine), hematuria, proteinuria, or structural renal disease (e.g. cysts).

Most patients with CKD also have hypertension secondary to that disease. This is particularly common in glomerular diseases, in which increased tubular sodium reabsorption expands the blood volume and raises cardiac output. The increased renal sodium reabsorption is likely due to tubular resistance to natriuretic peptides or due to primary increases in sodium transport activity.

Some renal disorders may cause low renal perfusion, leading to activation of the RAAS. Increased sympathetic nervous system activity can also be seen in CKD.

Hypertension in CKD should be treated with angiotensin-converting enzyme (ACE) inhibitors or angiotensin-receptor blockers (ARBs), as they slow the progression of disease.

Renal Artery Stenosis (RAS)

Renal artery stenosis (RAS), or narrowing of the renal artery, causes reduced renal blood flow. Since the kidneys sense blood pressure (afferent arteriole), a decrease in blood pressure and flow will trigger RAAS activation. The increased angiotensin-II and aldosterone levels act to raise the blood pressure.

In older patients, stenosis of the renal artery is usually associated with **atherosclerosis**. Patients may present with atherosclerotic risk factors (e.g. diabetes, hyperlipidemia, or cigarette use) and/or with other pre-existing atherosclerotic associated disease (e.g. myocardial infarction, stroke, peripheral artery disease, or mesenteric ischemia).

In contrast, with young patients, especially females without risk factors for atherosclerosis, consider **fibromuscular dysplasia**, which is a condition characterized by an c The condition may affect not just renal arteries but the internal carotid, vertebral, external iliac, and various visceral arteries.

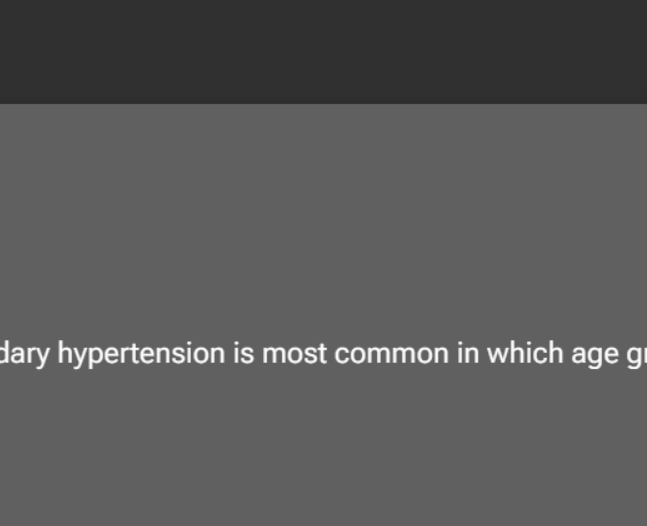

Figure 2. Fibromuscular Dysplasia. Note the thickening of arterial smooth muscle produces luminal stenosis. The characteristic appearance of the artery is described as a "string of beads."

Whether caused by atherosclerosis or fibromuscular dysplasia, renal artery stenosis results in a reduction of blood flow to the kidneys, activating the RAAS and leading to increases in blood volume and peripheral resistance, which results in secondary hypertension.

RAS should be suspected in newly hypertensive patients with underlying atherosclerotic disease. **Abdominal bruits** are heard in approximately 40% of patients and, if present, are specific for underlying RAS. Laboratory tests may show hypokalemia in 50% of patients due to high aldosterone levels, which cause potassium loss ific, however. Patients may also have elevated serum creatinine due to diminished blood flow to the kidneys, causing prerenal acute kidney injury.

The diagnosis of RAS can be confirmed with a Doppler ultrasound, nuclear medicine scan, or computed tomography or magnetic resonance angiography (Figures 3 and 4). A common feature of RAS on imaging studies is the presence of a single small kidney, reflecting unilateral involvement of either the right or left renal artery.

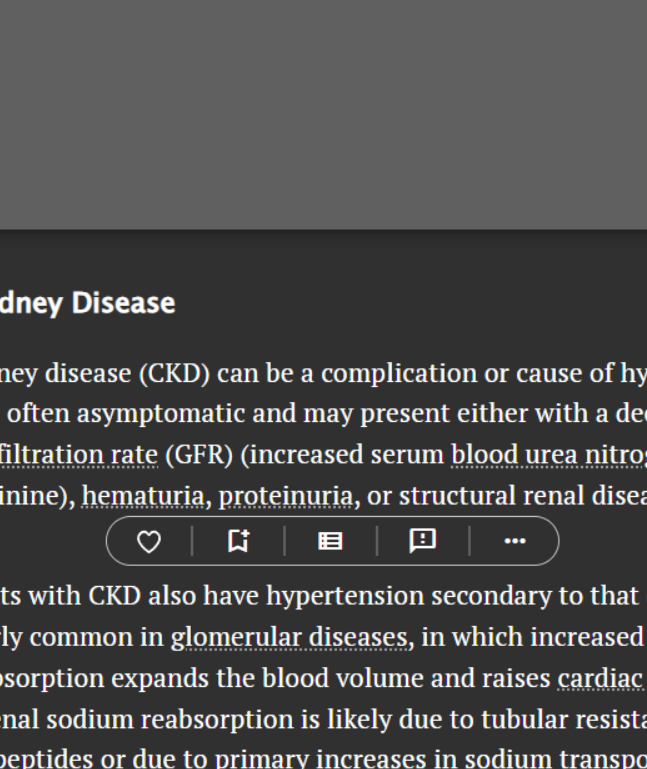

Figure 3. Right renal artery stenosis. This magnetic resonance angiography (MRA) image demonstrates atherosclerotic occlusion of the right renal artery (arrow) in a patient with diabetes; note the lack of contrast at the branch point of the renal artery from the aorta.

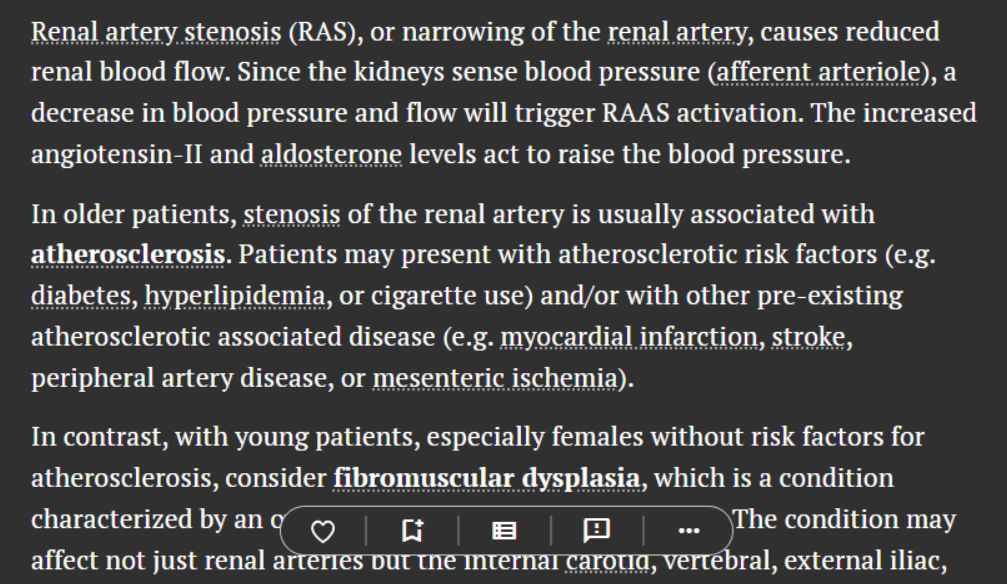

Figure 4. Fibromuscular Dysplasia. This renal artery angiogram demonstrates the characteristic "string of beads" appearance of fibromuscular dysplasia.

RAS is primarily treated medically; however, angioplasty and stenting are additional options which tend to be more effective in cases of fibromuscular dysplasia and in patients with severe symptoms.

For patients who cannot undergo angioplasty, antihypertensive agents are used to control blood pressure. The main goal of treatment is to preserve renal function. Patients with bilateral RAS may have an increase in serum creatinine (prerenal acute kidne Bs. This is because these drugs preferentially dilate the efferent arteriole, and, in the face of reduced renal perfusion, the glomerular capillary hydrostatic pressure and GFR can drop.

What Are Endocrine Causes of Secondary Hypertension? (LO #1)

The homeostatic regulation of blood pressure is a complex interplay between the cardiovascular, renal, nervous, and endocrine systems. The endocrine organs- adrenal and thyroid glands- are central to this endeavor. Since the hormones from these glands primarily increase blood pressure, secondary hypertension commonly results from excess hormone production.

Hyperaldosteronism

Suspect primary hyperaldosteronism in the setting of **unexplained hypokalemia** along with **new hypertension**. Primary hyperaldosteronism is usually caused by either an aldosterone-producing adrenal adenoma or idiopathic bilateral adrenal hyperplasia. The elevated aldosterone stimulates sodium and water reabsorption in the collecting ducts of the kidneys, resulting in elevated blood volume, which increases cardiac output and blood pressure. Aldosterone also promotes potassium excretion in the kidneys, resulting in hypokalemia. Correction of the high aldosterone levels by medications or surgery corrects the hypertension.

Why does hyperaldosteronism cause hypertension?

CLINICAL CORRELATION

Primary aldosteronism, also known as **Conn Syndrome**, is the overproduction of the hormone aldosterone by the adrenal glands. The key clinical feature of primary aldosteronism is **hypokalemia** in association with **hypertension**.

Hypercortisolism (Cushing Syndrome)

Patients with hypercortisolism will usually have one or more Cushingoid physical findings (Figure 5).

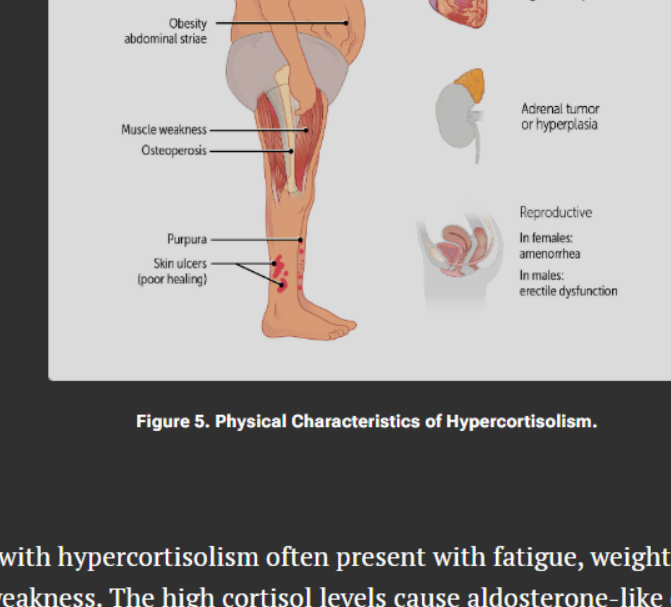
Pituitary tumor emotional disturbances
Moon face
Thin, wrinkled skin
Obesity abdominal striae
Cardiac hypertrophy high blood pressure
Muscle weakness
Osteoporosis
Adrenal tumor or hyperplasia
Purpura
Skin ulcers (poor healing)
Reproductive
In females: amenorrhea
In males: erectile dysfunction

Figure 5. Physical Characteristics of Hypercortisolism.

Patients with hypercortisolism often present with fatigue, weight gain, or muscle weakness. The high cortisol levels cause aldosterone-like effects by increasing sodium and water retention to increase blood volume. Increased expression of angiotensin-II receptors, increased effects of catecholamines on adrenergic receptors, and reduced nitric oxide (vasodilator) release combine to increase **systemic vascular resistance (SVR)**. Correction of the high cortisol levels by medications or surgery corrects the hypertension.

Thyroid Disorders

A change in thyroid hormone levels can alter the sensitivity of adrenergic receptors to catecholamines. Interestingly, both hyper- and hypothyroidism can cause secondary h s β-receptors sensitive to catechola ate, cardiac output, and blood pressure. Hypothyroidism may increase diastolic blood pressure due to constriction of peripheral vessels mediated by unopposed stimulation of α-receptors.

Pheochromocytoma

Pheochromocytoma is a **rare** catecholamine-secreting tumor, which may present with episodic signs of sympathetic overdrive like headaches, sweating, tremors, flushing, agitation, and tachycardia. Orthostatic hypotension is often seen, despite the basal hypertension. The excess catecholamines raise the blood pressure by increasing cardiac output and systemic vascular resistance (SVR). Correction of the high catecholamine levels by medications or surgery corrects the hypertension.

Congenital Adrenal Disorders

Some forms of congenital adrenal hyperplasia cause high cortisol and/or mineralocorticoids, so they can present with hypertension in children. These include 11β-hydroxylase and 17α-hydroxylase deficiencies, which also have high mineralocorticoids and hypokalemia.

CLINICAL CORRELATION

Congenital adrenal hyperplasia is a group of inherited disorders caused by mutations that disrupt the enzymes needed by the adrenal gland to produce hormones like cortisol, aldosterone, and androgens. These enzyme deficiencies lead to imbalances, both underproduction of some hormones and overproduction of others. Overprodu e, can cause the body to retain sodium and thus water, leading to increased blood volume and consequently high blood pressure.

How Does Sleep Apnea Cause Secondary Hypertension? (LO #1)

Obstructive sleep apnea (OSA) is a common cause of secondary hypertension. It is most often seen in people who are obese, but can also occur if there are airway abnormalities. A patient with OSA may have loud snoring at night, daytime fatigue, depression, and excessive daytime sleepiness.

Pathophysiology

In patients with OSA, the pharyngeal muscles and excess connective tissue in the neck collapse during sleep (Figure 6). This compresses their airway, momentarily stops breathing, and activates the sympathetic nervous system. Increased sympathetic tone increases cardiac output and peripheral resistance, which leads to hypertension. There is also an increased risk of developing pulmonary hypertension, as the chronic hypoxia can cause vasoconstriction of the blood vessels in the lungs. Diagnosis is by polysomnography. Treatment of OSA by continuous positive airway pressure (CPAP) slightly lowers the blood pressure, but medications are usually required to achieve normal blood pressure levels.

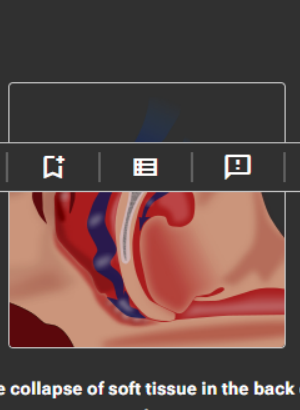

Figure 6. Obstructive Sleep Apnea. The collapse of soft tissue in the back of the throat obstructs the passage of air.

Why do patients with OSA have hypertension?

How Can Medications Cause Secondary Hypertension? (LO #1)

Hypertension is a side effect of certain medications that directly or indirectly disrupt the normal blood pressure regulatory pathway.

Pathophysiology

Many classes of drugs can cause secondary hypertension via various drug-specific mechanisms:

- **Oral contraceptive pills** that contain estrogen and estrogen replacement therapy can increase blood pressure because estrogen induces hepatic production of angiotensinogen. Angiotensinogen is converted to angiotensin-II, which directly causes hypertension through vasoconstriction and indirectly through aldosterone-mediated sodium and water retention.
- **Nonsteroidal anti-inflammatory drugs (NSAIDs)** can cause hypertension due to cyclooxygenase-2 (COX-2) inhibition in the kidneys. COX-2 inhibition decreases vasodilating prostaglandins that normally counteract the vasoconstricting effects of endogenous vasopressors such as angiotensin-II. This decreases sodium and water filtration and activates the RAAS.
- **Pseudoephedrine**, an α-agonist used as a decongestant, can cause dose-related increases in blood pressure through α-receptor mediated vasoconstriction. α-Agonists have a small effect on blood pressure (usually <5 mm Hg).
- **Cocaine and amphetamines** can cause secondary hypertension via α-receptor-induced vasoconstriction.
- **Chronic systemic glucocorticoid use** can also induce hypertension by increasing sodium and water retention and potentiating the vasoconstrictive effect of catecholamine (see discussion of Cushing syndrome above).

How do drugs like pseudoephedrine, cocaine, and amphetamines cause hypertension?

Table 1 summarizes some common causes of secondary hypertension in order of coverage.

Table 1

Cause	Pathophysiology	Diagnostic Clues
Chronic kidney disease (CKD)	↑ Tubular Na reabsorption or activated renin-angiotensin-aldosterone system (RAAS)	↑ Blood urea nitrogen (BUN) and creatinine, proteinuria, hematuria
Renal artery stenosis	↓ Renal blood flow activates RAAS	Abdominal bruit; history of atherosclerosis
Hyperaldosteronism	↑ Tubular sodium reabsorption	Hypokalemia
Hypercortisolism	↑ Tubular sodium reabsorption upregulates angiotensin II of catecholamines	Cushingoid physical findings; ure 24-hour urine cortisol
Thyroid disorders	Hyperthyroidism increases cardiac output Hypothyroidism increases vascular resistance	Hyperthyroid: increased pulse, weight loss Hypothyroid: depression, constipation, weight gain Measure thyroid stimulating hormone
Pheochromocytoma	↑ Catecholamines	Episodic palpitations, tremor; measure plasma or 24-hour urine metanephrines
Congenital adrenal hyperplasia	↑ Mineralocorticoids, ↑ tubular sodium reabsorption	Children, reproductive or pubertal abnormalities
Obstructive sleep apnea	Hypoxia increases sympathetic tone, which increases systemic vascular resistance (SVR)	Obese patients with excess neck soft tissue, snoring, daytime fatigue; perform polysomnography

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to JK, what is the most likely cause of her hypertension? How do you explain the diagnosis and management?

You suspect renal artery stenosis because of JK's resistant hypertension while taking multiple blood pressure medications and the presence of an abdominal bruit. A Doppler ultrasound shows a stenotic right renal artery and diminished renal blood flow. You explain to JK that her high blood pressure is likely due to narrowing of one of the arteries to her kidneys. You discuss treatment with renal artery angioplasty, but JK does not want "surgery" and will continue with medications for now.

Summary

- Secondary hypertension is elevated blood pressure due to an identifiable cause.
- Secondary hypertension accounts for 5% to 10% of all hypertension.
- Patients with chronic kidney disease (CKD) often have hypertension and may present with edema, proteinuria, or hematuria and elevated creatinine.
- CKD causes hypertension by increased tubular sodium reabsorption or activated renin-angiotensin-aldosterone system (RAAS).
- Renal artery stenosis (RAS) may present with a flank or abdominal bruit.
- RAS is caused by atherosclerosis in elderly patients with atherosclerotic risk factors; fibromuscular dysplasia is implicated in young females with no atherosclerotic risk factors.
- In RAS, decreased renal blood flow increases RAAS activity.
- Diagnosis of RAS is by MR angiography, Doppler, or nuclear scans.
- Treatment of RAS is with antihypertensive agents or sometimes angioplasty.
- Hyperaldosteronism can present with hypokalemia and causes hypertension via increased tubular sodium reabsorption.
- Hypercortisolism presents with features of Cushing syndrome (eg, moon facies, truncal obesity, striae) and causes hypertension via increased tubular sodium reabsorption and systemic vascular resistance (SVR).
- Hypertension due to thyroid dysfunction can occur with either hypothyroidism or hyperthyroidism.
- Pheochromocytomas present with episodic hypertension, orthostatic drop in blood pressure ; the high catecholamines increase SVR.
- Obstructive sleep apnea (OSA) can present with snoring and daytime sleepiness and causes hypertension by increasing sympathetic nervous system activity.
- Continuous positive airway pressure (CPAP) treatment will modestly decrease blood pressure.
- Hypertension is a side effect of certain medications that directly or indirectly disrupt the normal blood pressure regulatory pathway.
- Drugs associated with secondary hypertension include oral contraceptives, NSAIDs, decongestants, cocaine, methamphetamines, and corticosteroids.

Review Questions

1. A 27-year-old healthy female presents to the clinic for evaluation of new-onset hypertension. She is found to have an abdominal bruit on physical exam. Which of the following causes of hypertension is most likely in this patient?

A. ○ Atherosclerosis of the renal artery
B. ○ Fibromuscular dysplasia of the renal artery
C. ○ Obstructive sleep apnea
D. ○ Oral contraceptive usage
E. ○ Pheochromocytoma

Explanation (requires correct answer)

2. A 60-year-old obese male presents to the clinic with his wife. He is very tired, and his wife says that he falls asleep at the dinner table and snores loudly at night. His blood pressure is 145/95 mm Hg. Serum electrolytes, blood urea nitrogen, creatinine, and urinalysis are normal. Which of the following pathophysiologic mechanisms best explains his hypertension?

A. ○ Decreased glomerular filtration rate (GFR)
B. ○ Decreased vascular compliance
C. ○ Increased tubular sodium reabsorption
D. ○ Renin-angiotensin-aldosterone system (RAAS) activation
E. ○ Sympathetic nervous system (SNS) activation

Explanation (requires correct answer)

3. A 24-year-old female presents to the clinic for an annual physical exam. She feels well and a review of systems is negative except for nasal congestion and longstanding knee pain, and she has no family history of hypertension. Review of medications reveals that she takes an oral contraceptive, ibu daily pseudoephedrine for nasal congestion. Her blood pressure is 141/85 mm Hg. Her BMI is 22. All lab tests are normal. What is the most appropriate next step in management?

A. ○ Abdominal CT
B. ○ Enalapril
C. ○ Stop pseudoephedrine
D. ○ Urine metanephrines
E. ○ Weight-loss counseling

Explanation (requires correct answer)

4. A patient with elevated blood pressure and an audible abdominal bruit on physical exam most likely has secondary hypertension related to a pathologic condition involving which organ?

A. ○ Liver
B. ○ Thyroid
C. ○ Pancreas
D. ○ Heart
E. ○ Kidney

Explanation (requires correct answer)

5. A patient presents with complaints of persistent headaches, sweating, tremors, flushing and agitation. The patient is hypertensive while seated, but has a significant drop in blood pressure when he stands. CT imaging detects a mass in the patient's left adrenal gland. The cause of the patient's hypertensive is most likely overproduction of which type of hormone?

A. ○ Estrogen
B. ○ Insulin
C. ○ Catecholamines
D. ○ Vasopressin
E. ○ Growth hormone

Explanation (requires correct answer)

How would you rate this Brick?

References

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

- Bakris GL, Sorrentino MJ, eds. *Hypertension: A Companion to Braunwald's Heart Disease*. 3rd ed. Elsevier; 2018.
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Check off the objectives when you feel comfortable with the concept

01 ☐ Differentiate the clinical presentation, pathophysiology, diagnosis, and management of secondary hypertension caused by the various mentioned disorders.

Objective Confidence: 0/1